

Mano Bhavdeep Venkatesan

Schwenningen, Germany | Master's Graduate - Hochschule Furtwangen University

Biomedical Engineer | MEMS & Microsystems Research | Microfabrication & Experimental Characterization | Wearable Medical Technologies

Biomedical Engineer with an M.Sc. in Biomedical Engineering and research experience in MEMS, 3D microsystems, thin-film materials and experimental characterization. Master's research at Hahn-Schickard focused on the development and characterization of 3D acoustic MEMS membranes using FT-IR spectroscopy, supported by XRD, SEM, wafer-bow analysis and MATLAB-based modelling. Experienced in independently planning and executing experimental research, developing characterization methodologies, analysing multimodal datasets and translating measurements into material and device-level understanding. Earlier experience in biomedical engineering includes medical-device installation, validation, maintenance and quality workflows across clinical environments. Interested in advancing this multidisciplinary foundation toward microfluidics, BioMEMS, soft materials and wearable medical microsystems for non-invasive healthcare.

CAREER TIMELINE

Hochschule Furtwangen University

Sep 2022 – Oct 2025

Master of Science – Biomedical Engineering

- Completed coursework in biomedical modelling, physiological simulation, medical imaging, and machine learning applications.

SIMATS (Saveetha University), Chennai, India

Jan 2019 – Jan 2022

Biomedical Engineer

- Installed, operated, and maintained SEM and INSTRON tensile/compression testers for routine failure analysis and mechanical characterization of medical device components and biomaterials, ensuring 99% lab uptime in a high-throughput clinical environment.
- Implemented QA workflows aligned with NABH and ISO-based safety standards, reducing equipment audit non-conformities by improving calibration, documentation, and preventive maintenance routines.
- Installed, calibrated, and validated >50 diagnostic and monitoring systems (imaging, cardiology, and critical-care devices), ensuring compliance with performance specifications and safety protocols.

Madras Medical Mission, Chennai, India

Aug 2018

Internship

- Rotated through cardiology, ICU, and imaging departments; shadowed biomedical engineers in safety checks, basic troubleshooting, and documentation for diverse medical devices. Gained hands-on exposure to hospital compliance workflows, equipment operation, and preventive maintenance practices.

Jerusalem College of Engineering – Anna University

Jun 2014 – May 2018

Bachelor of Engineering - Biomedical Engineering

- Completed undergraduate degree in Biomedical Engineering with focus on medical imaging, biomedical instrumentation, and signal processing.

ACADEMIC PROJECTS

❖ M.Sc. Master Thesis - Hahn Schickard Geschellschaft

April 2025 – Oct 2025

Master Thesis Researcher

Topic: Development and characterization of 3D membranes using FT-IR Spectroscopy for acoustic applications | Grade 1.0

- Developed and experimentally characterized 3D acoustic MEMS membranes with AIN thin films, addressing characterization challenges associated with non-planar microsystem structures.
- Designed and implemented FT-IR characterization protocols for complex 3D structures, including reflection-mode measurements and sidewall characterization.
- Correlated FT-IR measurements with XRD, SEM and wafer-bow analysis to evaluate thin-film crystallinity, orientation, stress and process-dependent material behaviour.
- Developed MATLAB-based Fresnel optical models and phonon-fitting workflows for quantitative interpretation of FT-IR data.
- Independently planned experiments, evaluated multimodal datasets and developed reproducible measurement procedures from experimental concept through methodology validation.
- Presented the research as part of the accepted EUROSensors 2026 conference contribution.

❖ GTT Mathematical Modelling – Glucose Dynamics (Simulation in Biomedicine) Wise 2022 – 2023

Fitted damped oscillatory model to 2 subject's glucose data; extracted physiological decay & oscillation parameters - relevant to diabetes diagnostics & wearable glucose monitoring

❖ Detection of Out-of-Body Images in Laparoscopic Videos - Research Project

Sose 2023

Implemented MATLAB pipelines for detecting out-of-body frames in laparoscopic videos using supervised classifiers (SVM, decision trees, neural network models) on features derived from colour histograms, motion cues, and specular highlights. Evaluated models on labelled surgical datasets to improve tool-in-body frame detection and reduce surgeon distraction.

- ❖ **Alveoli Dynamics – Hickling Model (Physiological Modelling)** Sose 2023
Optimized PEEP ventilation levels; modelled cardiac effects on alveolar recruitment - applicable to ventilator design & critical care device development
- ❖ **Bachelor Thesis - Automated Detection of Maculopathy in SD-OCT images** Jan 2018 – May 2018●
Developed an image-processing pipeline for SD-OCT retinal images incorporating PM3D denoising, retinal-layer segmentation, feature extraction and classification. Investigated automated classification of retinal conditions using multilayer SVM classifiers and training datasets. Applied quantitative image-analysis methods to extract clinically relevant features from retinal OCT data.

PUBLICATIONS

- **Conference Poster**
3D acoustic MEMS membrane characterization using FT-IR spectroscopy
Based on Master's thesis research at Hahn-Schickard; conference presentation/publication in progress.

ADDITIONAL GERMAN INDUSTRY EXPERIENCE

- ❖ **Celebrate GmbH - Production Assistant** June 2026 – Present
- ❖ **Celebrate GmbH - Seasonal Production Employment - Production Assistant** Nov-Dec [2024, 2025]
- ❖ **Eisenmann Druckguss GmbH - Production Employee** 2 months [2023]

TECHNICAL SKILLS

Category	Skills
MEMS & Microsystems	3D MEMS structures, acoustic membranes, microsystem characterization, thin-film integration, device-level experimental analysis
Materials & Characterization:	FT-IR spectroscopy, XRD, SEM, wafer-bow analysis, AIN thin films, crystallinity, crystal orientation, residual stress, morphology evaluation
Experimental Research:	Experimental planning, characterization-method development, protocol development, multimodal data correlation, scientific analysis, troubleshooting
Modelling & Data Analysis:	MATLAB, Fresnel optical modelling, Voigt phonon fitting, signal processing, image analysis, LSQNONLIN optimization, quantitative data analysis
Biomedical Engineering:	Medical-device engineering, biomedical instrumentation, medical imaging, diagnostic technologies, biosignal/image analysis, healthcare technology

INTERESTS

MEMS sensors and acoustic/ultrasonic systems | Bio-interface engineering and osseointegration | Smart Implants & Bio-Interfaces | Materials characterization and surface analytic | R&D from lab to product.

CERTIFICATES

Medical Device Regulation (EU MDR) - Udemy | Python Fundamentals – Udemy | Biomaterials for Bioprinting – Udemy | Six Sigma Green Belt - LinkedIn Learning course

LANGUAGES

DE German – B1
(Actively Improving)

GB English – Fluent (C1)

Tamil – Native

Date: September 2026 | Place: Villingen-Schwenningen

V.R. BH

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